

(d) Minimize $z = 5x_1 - 4x_2 + 6x_3 - 8x_4$.

*3. Consider the following system of equations:

$$\begin{aligned} x_1 + 2x_2 - 3x_3 + 5x_4 + x_5 &= 8 \\ 5x_1 - 2x_2 + 6x_4 + x_6 &= 16 \\ 2x_1 + 3x_2 - 2x_3 + 3x_4 + x_7 &= 6 \\ -x_1 + x_3 - 2x_4 + x_8 &= 0 \\ x_1, x_2, \dots, x_8 &\geq 0 \end{aligned}$$

Let $x_5, x_6, \dots$, and x_8 be a given initial basic feasible solution. Suppose that x_1 becomes basic. Which of the given basic variables must become nonbasic at zero level to guarantee that all the variables remain nonnegative, and what is the value of x_1 in the new solution? Repeat this procedure for x_2, x_3 , and x_4 .

4. Consider the following LP:

$$\text{Maximize } z = x_1$$

subject to

$$\begin{aligned} 5x_1 + x_2 &= 4 \\ 6x_1 + x_3 &= 8 \\ 3x_1 + x_4 &= 3 \\ x_1, x_2, x_3, x_4 &\geq 0 \end{aligned}$$

(a) Solve the problem *by inspection* (do not use the Gauss-Jordan row operations), and justify the answer in terms of the basic solutions of the simplex method.

(b) Repeat (a) assuming that the objective function calls for minimizing $z = x_1$.

Solve the following problem *by inspection*, and justify the method of solution in terms